

Analyzing and predicting user behavior in online marketing

Bachelor's Thesis – Torben Nattermann

Karlsruhe, 01.03.2024



Motivation



Context

Nvidia profitiert von KI-Boom – Chiphersteller zieht Börsen nach oben

Dank des Runs auf KI-Lösungen hat der Chipkonzern Nvidia starke Zahlen vorgelegt, der Dax stieg daraufhin auf einen neuen Höchstwert. KI-Entwickler Sam Altman warnt aber auch vor Risiken der Technologie.

22.02.2024, 09:31 Uhr

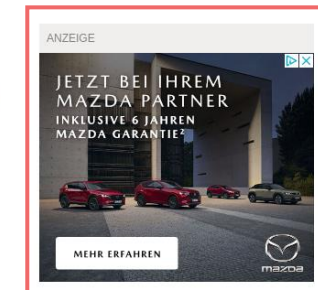


Nvidia-Logo: Wette auf eine durch KI geprägte Zukunft. Foto: Dado Ruvic / REUTERS

Ad

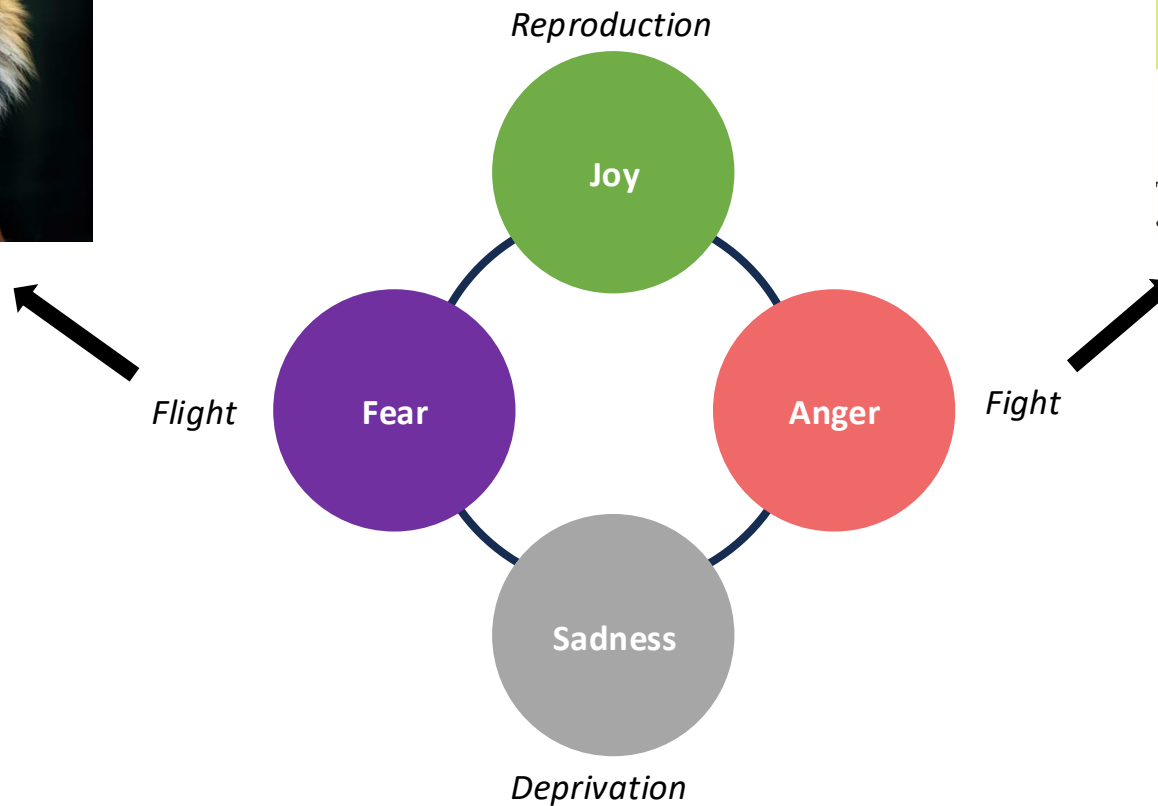
Der Boom beim Einsatz **künstlicher Intelligenz (KI)** sorgt weiterhin für explosives Wachstum beim **Chipkonzern Nvidia**. Im vergangenen Quartal lag der Umsatz mit 22,1 Milliarden Dollar mehr als dreimal so hoch wie ein Jahr zuvor. Analysten hatten im Schnitt mit 20,4 Milliarden Dollar gerechnet.

Die Aktie legte im nachbörslichen US-Handel um gut neun Prozent zu, der Dax stieg zu Handelsbeginn um 1,3 Prozent auf über 17.300 Punkte und übertraf damit seine jüngste Bestmarke, die er Ende vergangener Woche aufgestellt hatte. In **Japan** stieg der Leitindex Nikkei 225 nach Bekanntgabe der Nvidia-Zahlen so hoch wie seit rund 35 Jahren nicht mehr.



Fundamentals

Why **emotions**?





Related Work I

Paper	Modality	Prediction goal	Dataset	Language	Features	Model
Gupta & Yang (2018)	Text	Emotions	Twitter	English	FastText & TF-IDF embeddings + lexicon & linguistic features	SVM
Babanejad (2019)		User interest	Newspaper	English	TF-IDF embeddings + lexicon features	NN
Kheiri & Karimi (2023)		Sentiments	Twitter	English	ADA embeddings	SVM
Yang et al. (2017)	Image	Emotions	Affective images	-	VGG Net	NN
Klinger et al. (2022)	Text + Image	Emotions	Tumblr	English	Bert embeddings + ResNet	NN
Zhen (2022)		Sentiments	Twitter	English	Bert embeddings + ResNet	NN

 text  image  multimodal

RF = Random Forest,
SVM = Support Vector Machine,
NN = Neural Network



Related Work II: Comparison

Paper	Target	Metric	Image	Text	Multimodal
Klinger et al. (2022)	8 emotions	F1 score	0.41	0.53	0.53
Zhen et al. (2022)	3 sentiments	F1 score	0.61	0.66	0.69

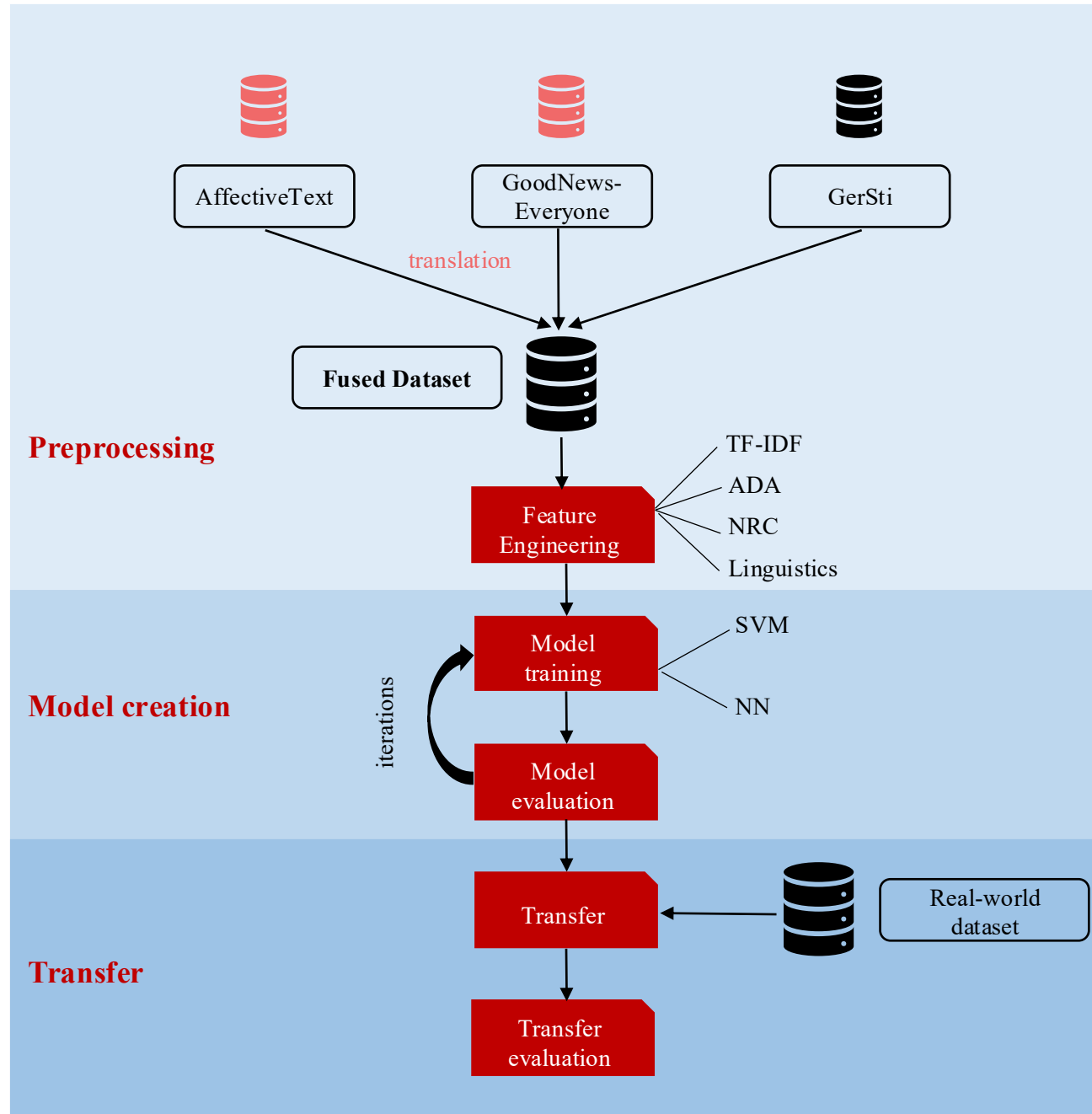
Focus on text:

- ❖ No significant performance increase for multimodal consideration
- ❖ Editorial contexts

Related Work III: Synthesis

Paper	Prediction goal	Modality	Dataset	Language	Features	Model
Gupta & Yang (2018)	Emotions	Text	Twitter	English	FastText & TF-IDF embeddings + lexicon & linguistic features	SVM
Babanejad (2019)	User interest	Text	Newspaper	English	TF-IDF embeddings + lexicon & linguistic features	NN
Kheiri & Karimi (2023)	Sentiments	Text	Twitter	English	ADA embeddings	RF & SVM
This work	Emotions	Text	Newspaper headlines	German	ADA embeddings + lexicon & linguistic features	SVM & NN

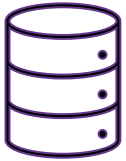
Concept



NRC = Emotion Lexicon

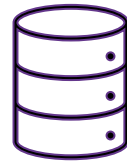
Preprocessing: Dataset

Combination of the 3 labeled **news headline emotion** datasets



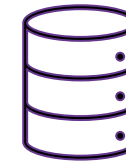
AffectiveText

- ❖ 1050 samples
- ❖ translated



GoodNewsEveryone

- ❖ 5000 samples
- ❖ translated

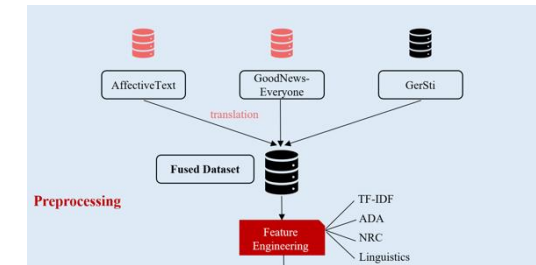


GerSti

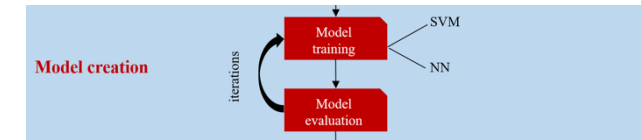
- ❖ 1000 samples
- ❖ German

Resulting merge

- ❖ **6353** samples for 11 emotions + neutral
- ❖ **3167** samples for 4 emotions + neutral



Model creation: Setup



Iterative experimentation to refine model and feature combinations

First iteration

❖ Isolated feature sets

Second iteration

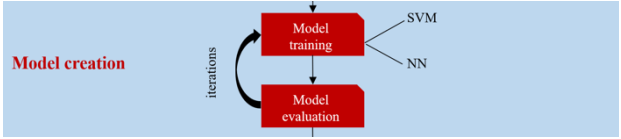
❖ Combined feature sets

Third iteration

❖ Hyperparameter tuning for best models

Model creation: Comparison

Iteration	Model	Base		Complete	
		Train F1	Val F1	Train F1	Val F1
1	SVM_Linguistics	0.246	0.248	0.110	0.110
1	SVM_TF-IDF	0.923 ↔	0.430	0.906 ↔	0.227
1	SVM_NRC	0.610	0.412	0.672	0.231
1	SVM_ADA	0.981 ↔	0.587	0.996 ↔	0.357

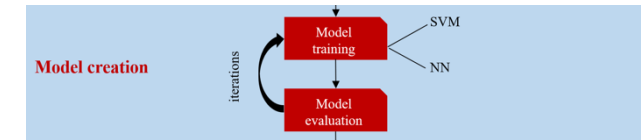


Overfitting

➡ ADA best feature

Model creation: Comparison

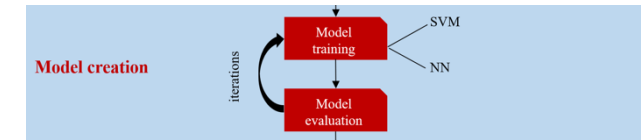
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1	SVM_ADA	0.981	0.587	0.996	0.357
2	SVM_TF-IDF+NRC	0.935	0.451	0.929	0.254
2	SVM_ADA+TF-IDF	0.985	0.600	0.997	0.343
2	SVM_ADA+NRC	0.980	0.601	0.996	0.357
2	SVM_ADA+TF-IDF+NRC	0.985	0.602	0.997	0.350



- **ADA** best feature
- **Baseline** implementation of Babanejad (2019)
- **ADA+NRC** performs best for both emotion sets

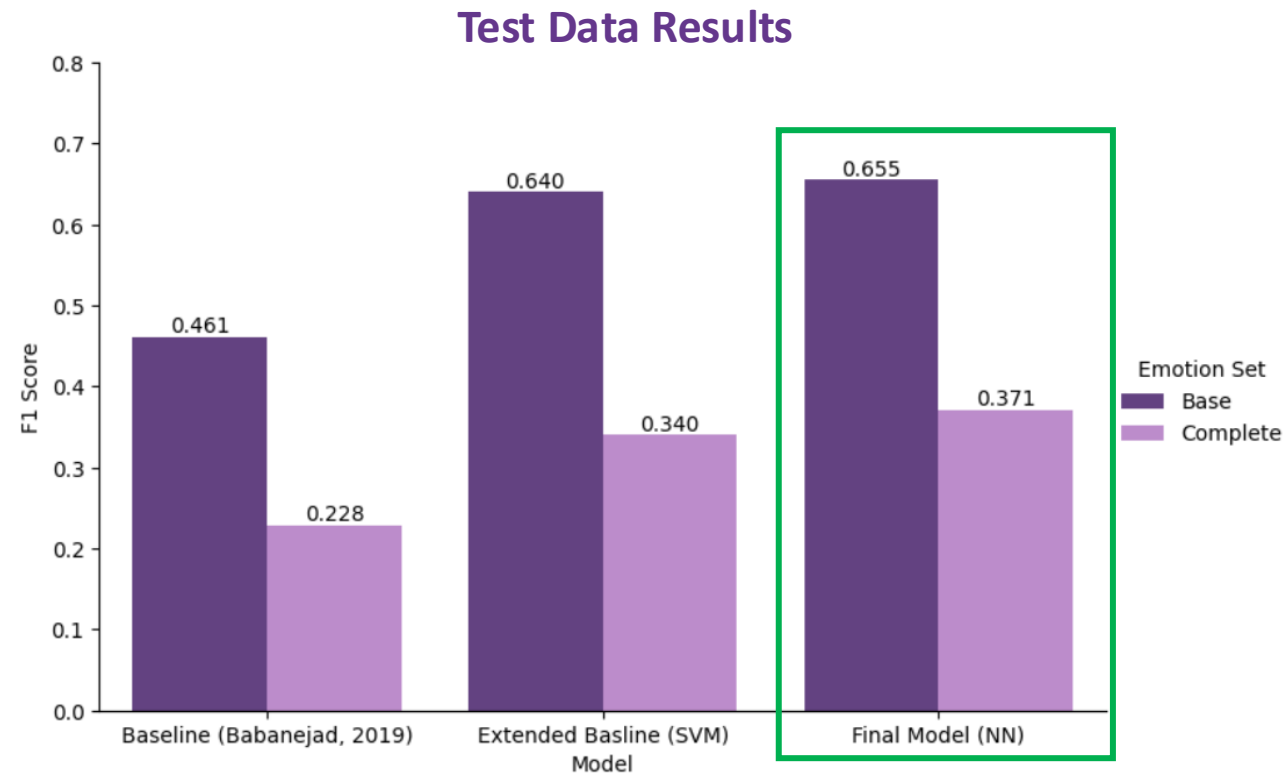
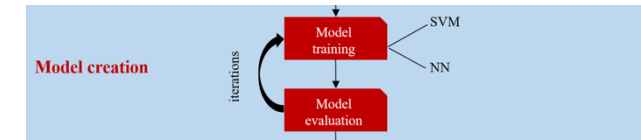
Model creation: Comparison

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2	SVM_ADA+NRC	0.980	0.601	0.996	0.357
2	SVM_ADA+TF-IDF+NRC	0.985	0.602	0.997	0.350
3	SVM_ADA+NRC tuned	0.980	0.602	0.996	0.357
3	NN_ADA+NRC tuned	0.820	0.617	0.998	0.357

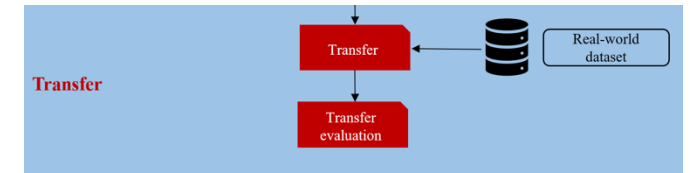


- **ADA** best feature
- **Baseline** implementation of Babanejad (2019)
- **ADA+NRC** performs best for both emotion sets
- **Neural network** performs best

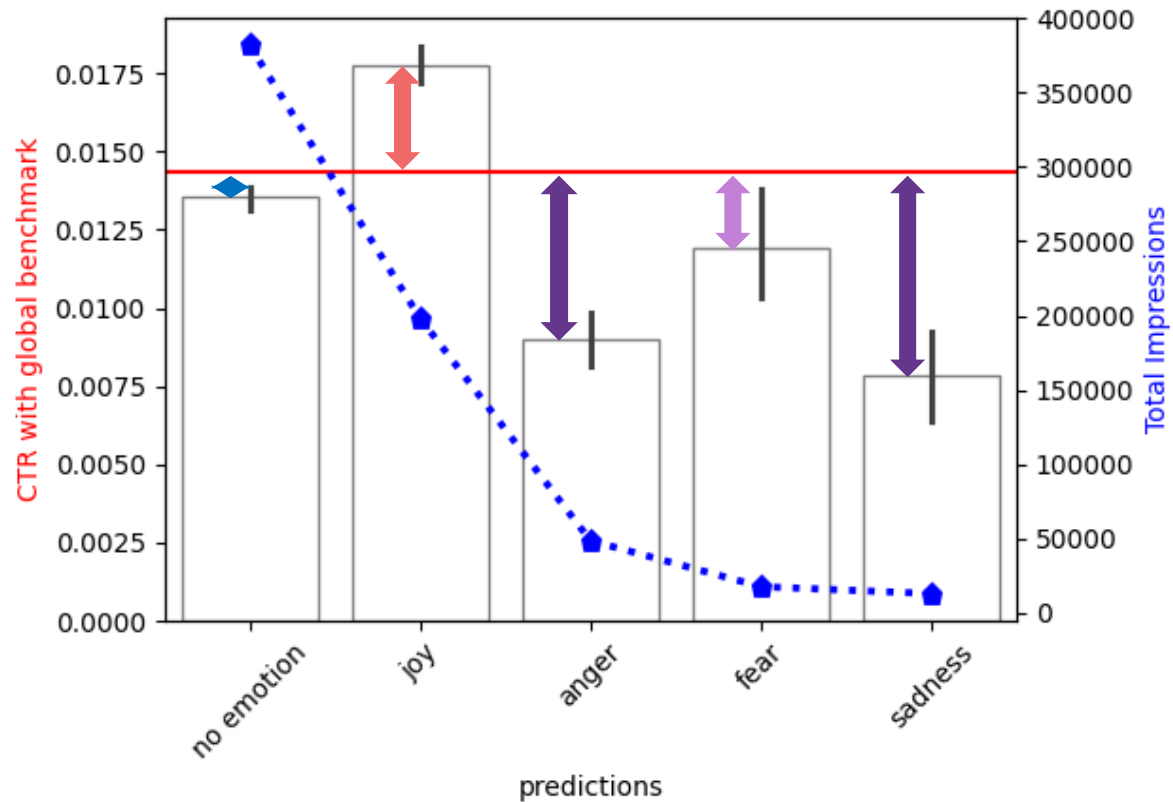
Model creation: Test Performance



Transfer: Behaviour prediction



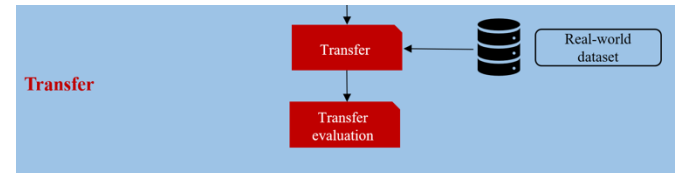
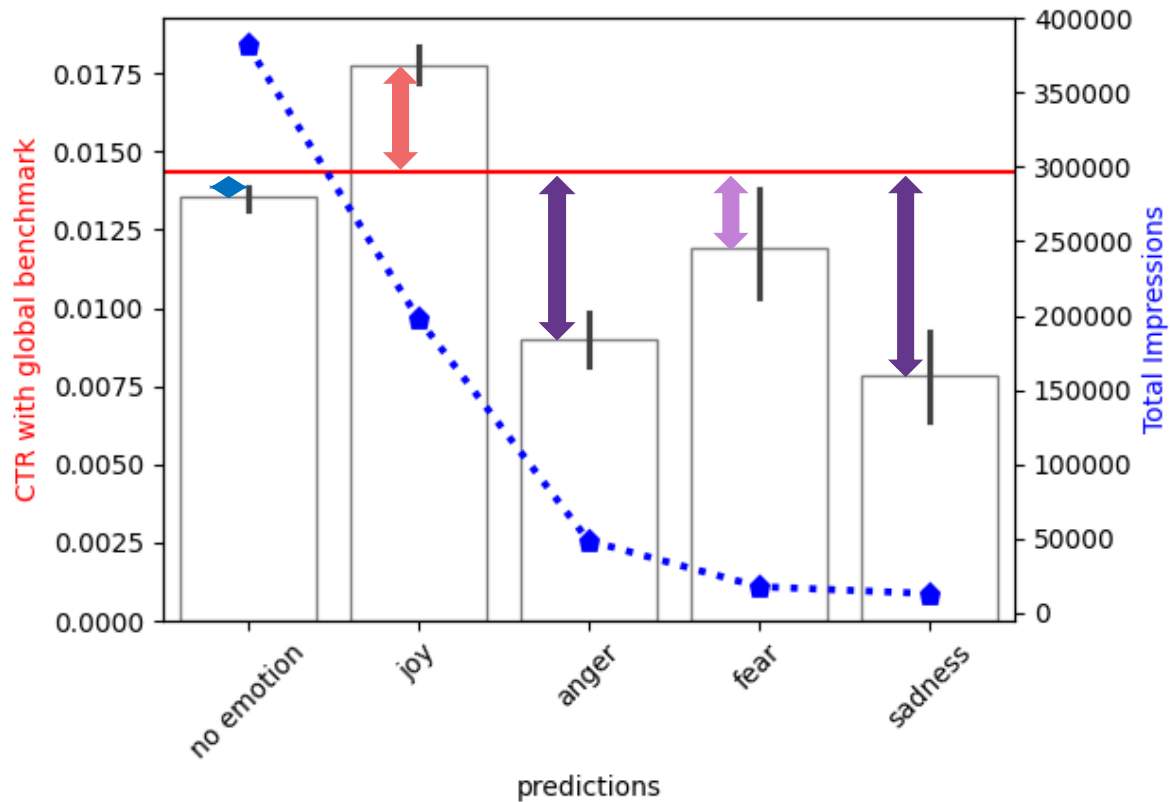
CTR and Total Impressions for Basic Emotions



- ❖ **No emotion** closest to global benchmark
- ❖ **Positive** emotions perform better than **negative** ones
- ❖ **Fear** performs best amongst negative emotions

Transfer: Behaviour prediction

CTR and Total Impressions for Basic Emotions



Eishockeyprofi wird von Schlittschuh getroffen und stirbt

Bei einer Partie in der englischen Eishockey-Liga soll der Spieler Adam Johnson von einer Kufe am Hals geschnitten worden sein. Sein deutsches Ex-Team Augsburger Panther nimmt Anteil an Johnsons Tod.

CTR: 0,75%

Einbrecher zerstören Werk aus Blei im Lager des Künstlers Anselm Kiefer

Unbekannte sind in das Lager Anselm Kiefers eingebrochen und haben ein monumentales Werk zerstört. Das ist nicht der erste Einbruch bei dem Künstler, doch um die Kunst ging es in der Regel nicht.

CTR: 0,91%

Du bist ständig müde und erschöpft?

Das kann an einem Vitamin-B12-Mangel liegen. Jeder Mensch benötigt Vitamin B12 für einen gesunden Stoffwechsel. Fällt der Wert unter einen bestimmten Bereich, entsteht ein Mangel mit typischen Symptomen wie Müdigkeit und Erschöpfung. Die Ursachen sind vielfältig.

[Mehr erfahren](#)

CTR: 1,76%

25 Meter hoher Weihnachtsbaum in Lingen beginnt zu glitzern

CTR: 1,2%

Supervulkan in Italien: Das droht, wenn er ausbricht

Conclusion

Summary



Text based **emotion extraction** →
baseline **outperformed**



Base **emotions** differentiate and
influence **user behaviour**



Class ambiguities and **language complexities**

Further Work



Integrate **further data** sources

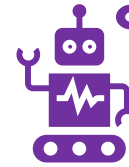


Train classifier on **ad** contents
→ **interaction** effects

Thank You for Your attention!

Human versus Machine:

What emotions do these headlines evoke in you?

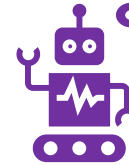


Anger

Die Nerven liegen blank bei der DFB-Elf: So wird das nichts bei der EM



Österreichs Nationalspieler-Veteran Marko Arnautovic begleitet Sané seelsorgerisch vom Platz
© Alexander Hassenstein / Getty Images



Sadness

Schmuggler sollen Geflüchtete von Schnellboot gedrängt haben – vier Tote

Mit Schnellbooten schaffen Schmuggler eigentlich Drogen aus Afrika nach Spanien. Inzwischen aber sollen sie immer öfter Menschen zur Flucht verhelfen – und gehen dabei offenbar skrupellos vor, wie ein aktueller Fall zeigt.

01.12.2023, 16.15 Uhr



Schnellboot vor der Küste von Cádiz (Bild von 2018) Foto: Emilio Moresatti / AP

+

K

A

Sources

<https://www.statista.com/outlook/amo/advertising/worldwide#ad-spending>

<https://socialanalyticsplus.net/crystalfeel/>

<https://digital-strategy.ec.europa.eu/de/policies/digital-services-act-package>

Otamendi, F. J., & Sutil Martín, D. L. (2020). The emotional effectiveness of advertisement. *Frontiers in psychology*, 11, 2088.

Plutchik, R. (1984): “Emotions: A general psychoevolutionary theory.” *Approaches to emotion 1984*: pp. 197–219.

SOURCES DATASETS

Image Sources

<https://www.vecteezy.com/free-photos/tiger-roar>

https://lp.thieme.de/is/cne-notfallpflege/course/1272/de_DE/innercontent.13778.3.2.html

<https://www.spiegel.de/wirtschaft/unternehmen/nvidia-profitiert-von-ki-boom-chiphersteller-zieht-dax-nach-oben-a-72c90dad-fd9d-440b-b59a-97532a249b47>

<https://www.stern.de/sport/fussball/dfb-elf--bei-der-nationalmannschaft-ist-besserung-kaum-in-sicht-34221556.html>

<https://www.spiegel.de/ausland/spanien-schmuggler-sollen-gefluechtete-von-schnellboot-gedraengt-haben-vier-tote-a-7e861d8f-5f77-4a37-ace9-17ad7d8b6b89>

<https://www.spiegel.de/sport/eishockey/adam-johnson-eishockey-profi-wird-von-schlittschuh-getroffen-und-stirbt-a-3cdc586e-aa26-42e0-9772-90d29d30ace6>

<https://www.spiegel.de/panorama/justiz/anselm-kiefer-einbrecher-zerstoeren-kunstwerk-in-seinem-lager-in-paris-a-d71d8c69-89ef-4304-9074-a3b7e98ac84>

<https://www.noz.de/lokales/lingen/artikel/25-meter-hoher-weihnachtsbaum-in-lingen-beginnt-zu-glitzern-20158461>

https://www.allgaeuer-zeitung.de/meistgesucht/vulkan-supervulkan-in-italien-das-droht-wenn-er-ausbricht-21-11-23_arid-658094

Hyperparameters

```
'parameters': {
  'hl1_size': {'values': [1024, 512]},
  'hl2_size': {'values': [1024, 512, 256]},
  'hl3_size': {'values': [128, 64]},
  'lr': {'values': [0.0001, 0.0005, 0.001]},
  'use_third_hidden_layer': {'values': [True, False]},
  'batch_size': {'values': [32, 64]}
}
```

```
{
  'estimator_C': [0.001, 0.01, 0.1, 0.5, 1, 15, 20, 50, 100],
  'estimator_kernel': ['linear', 'poly', 'rbf', 'sigmoid']
}
```

Base Emotions:

- Batch size: 64
- Learning rate: 0.001
- Optimizer: adam

HL1: 1566 X 1024 (relu)

HL2: 1024 X 256 (relu)

HL3: 256 X 64 (relu)

HL4: 64 X 5 (softmax)

Complete Emotions:

- Batch size: 64
- Learning rate: 0.0005
- Optimizer: adam

HL1: 1566 X 1024 (relu)

HL2: 1024 X 256 (relu)

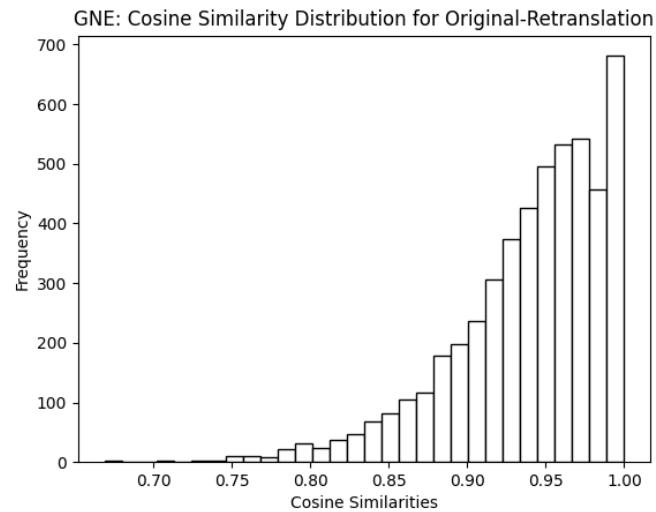
HL3: 256 X 12 (softmax)

Related Work III: Text Datasets

Name	Author	Taxonomy	Language	Sample type	Sample size
ISEAR	Scherer & Wallbott 1994	7 emotions	English	Emotion expressions	3000
SemEval2018	Mohammad et al. 2018	4 emotions	English	Tweets	11.000
EmoBank	Buechel & Hahn 2017	Valence, Arousal, Dominance	English	Emotion expressions	10.000
GoEmotions	Demszky et al. 2020	26 emotions + neutral	English	Reddit comments	58.000
CrowdFlower Data	CrowdFlower	13 emotions + neutral	English	Tweets	40.000
AffectiveText2007	Strapparava & Mihalcea 2007	6 emotions	English	News headlines	1050
GerSti	Dang et al. 2021	9 emotions + neutral	German	News headlines	1000
GoodNewsEveryone	Oberländer et al. (2020)	15 emotions	English	News headlines	5000

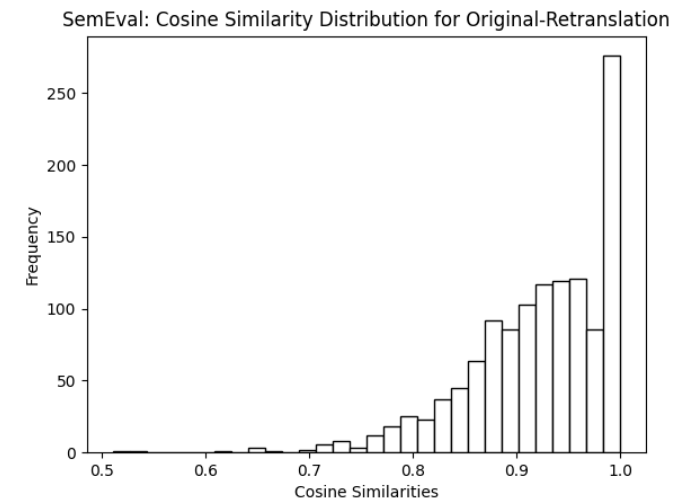
Translation Evaluation

GNE dataset



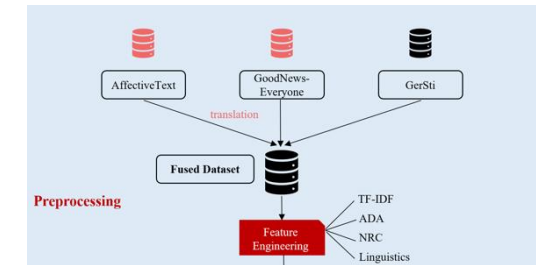
Mean cos_sim: 0.939
Threshold set to 0.85, resulting in a selection percentage of 94 % of cases
Mean cos_sim afterwards: 0.947

SemEval2007 dataset



Mean cos_sim: 0.921
Threshold set to 0.85, resulting in a selection percentage of 86,4 % of cases
Mean cos_sim afterwards: 0.941

Preprocessing: Feature Engineering



TF-IDF

- Weights for uni- & bigrams (Babanejad, 2019)
- **1000 features**

ADA

- ada002 embeddings (Kheiri & Karimi, 2023)
- **1536 features**

Lexica
(NRC)

- Cosine Similarities to class centroids and word counts (Gupta & Yang 2018; Babanejad 2019)
- **30 features**

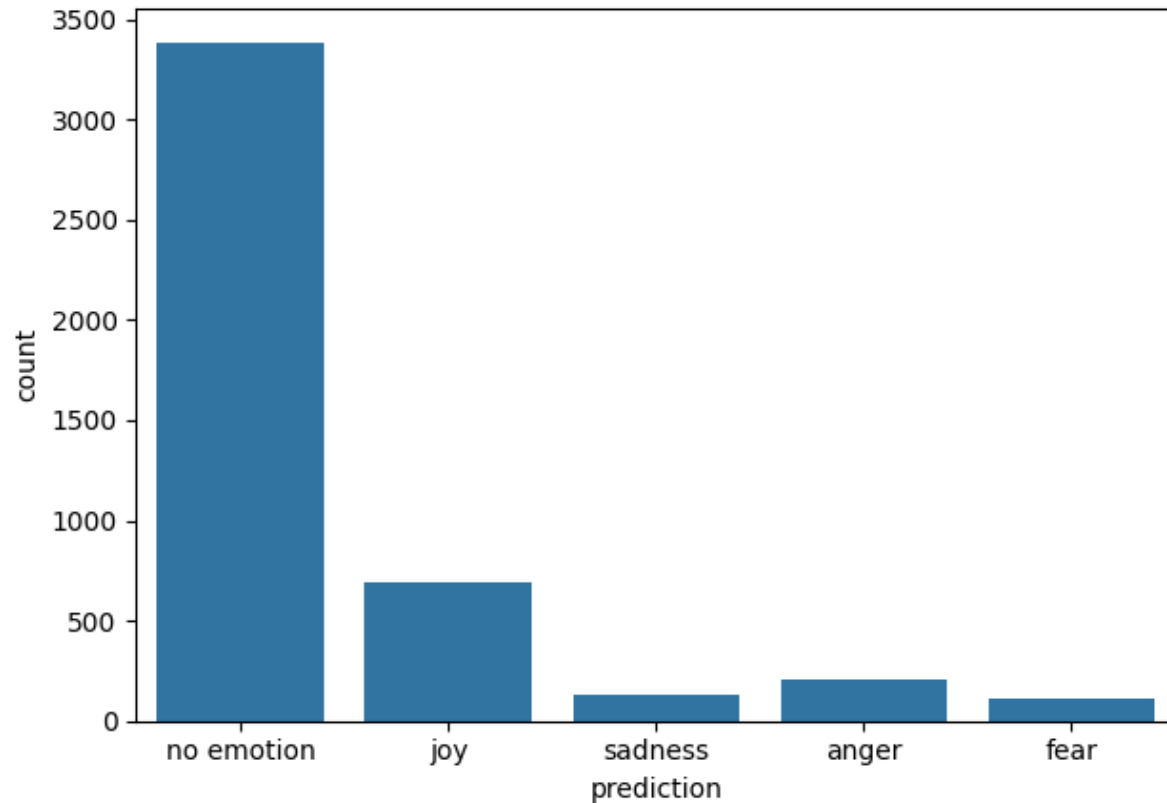
Linguistics

- Linguistic markers (Gupta & Yang 2018)
- **4 features**

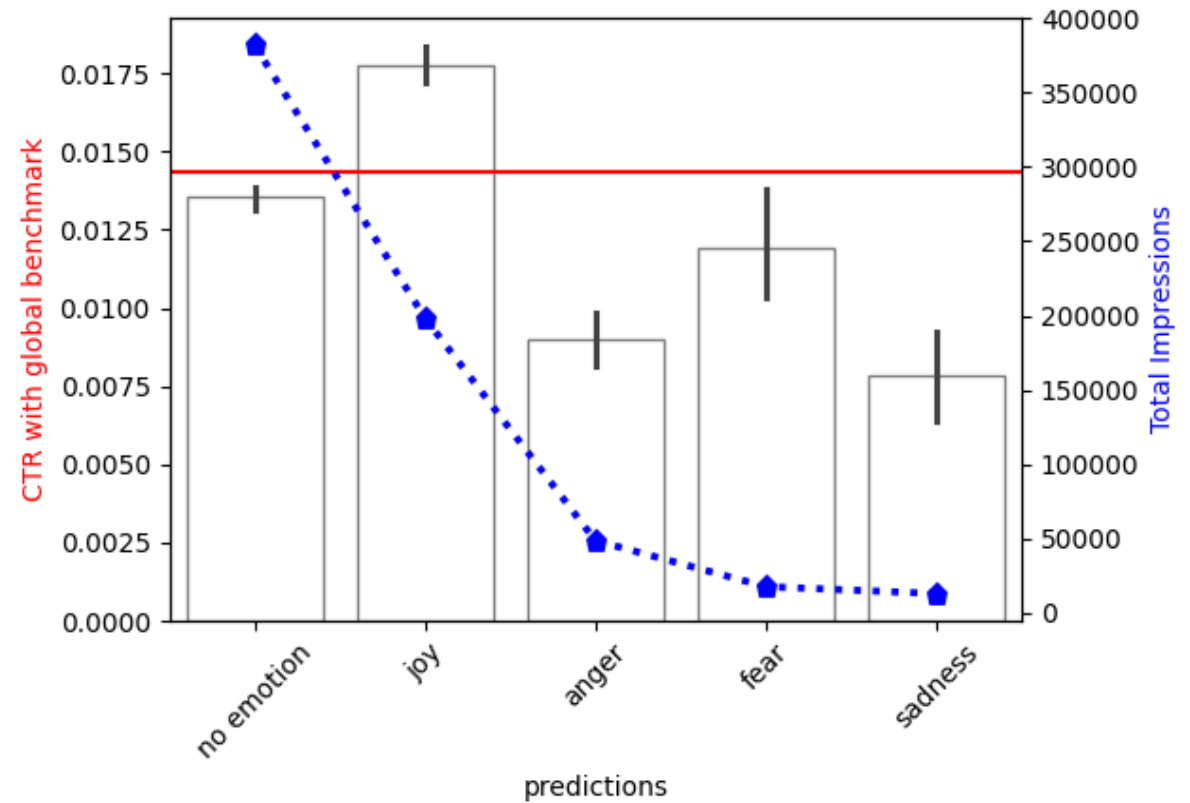


Transfer to main corpus: base

Predicted Class Distribution for Basic Emotions



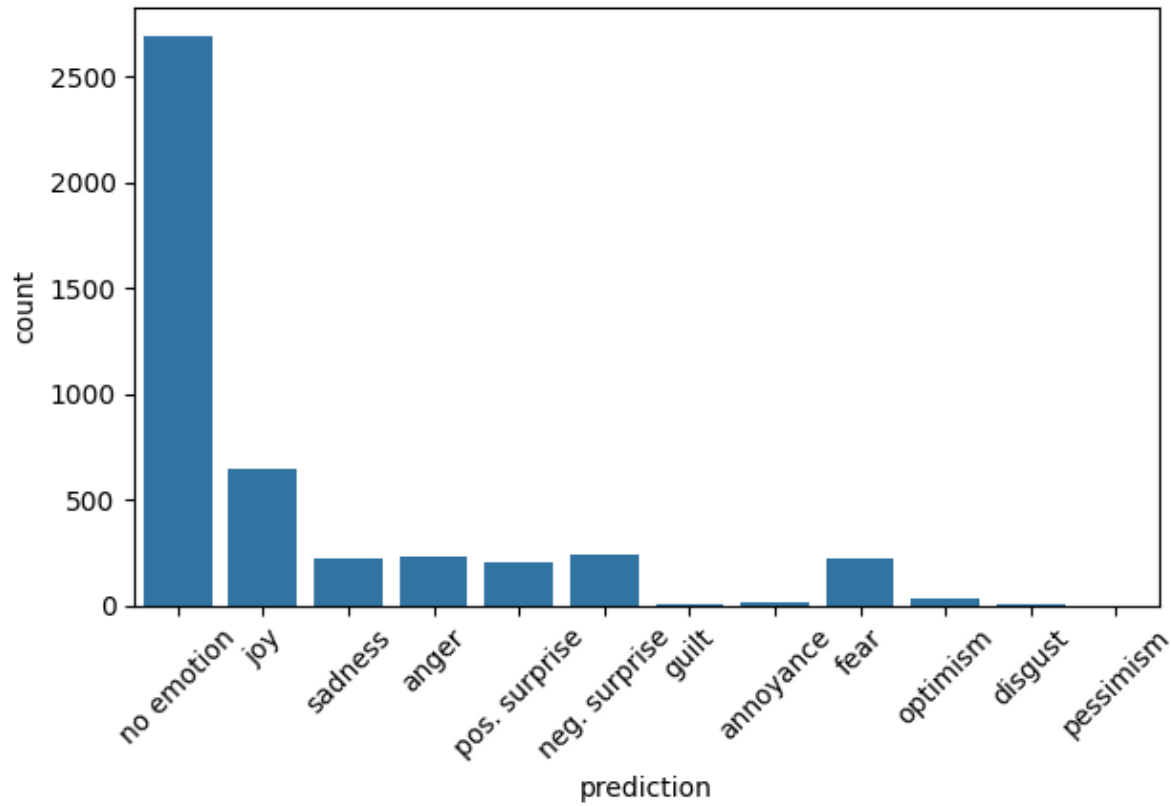
CTR and Total Impressions for Basic Emotions



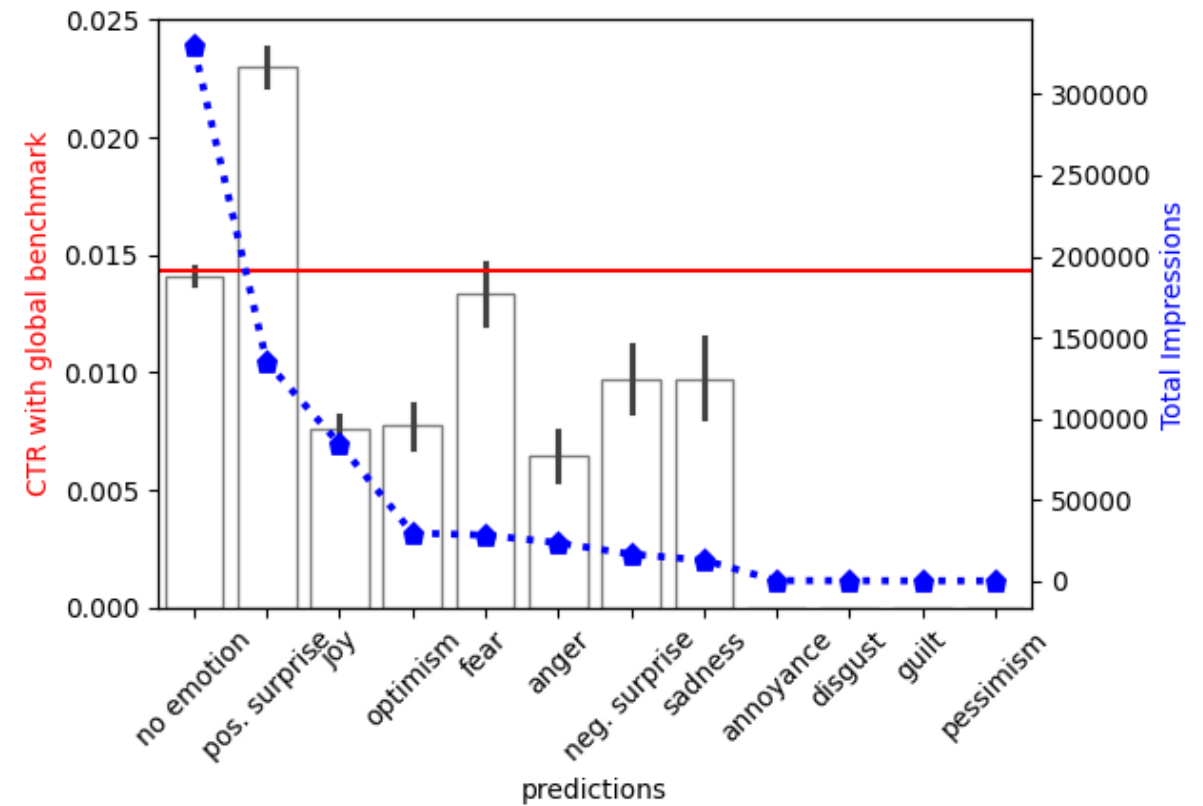


Transfer to main corpus: complete

Predicted Class Distribution for Complete Emotions



CTR and Total Impressions for Complete Emotions



Problem statement & Scope

How can **contextual information** be used to **predict user preference** and **behavior** for the online market?



Identification of **relevant concepts**



Systematic **information extraction**



Development of classification **algorithms**



Evaluation of **effectiveness** on real-world data

Further Work



Integrate **further data** sources (text & image)



Train classifier on **ad** contents → **interaction** effects



Improve **German data** sources and availability



Analyze patterns on **real-world data** (topic clustering)